

anyone with the knowledge of C/C++ to program hardware. The Galileo platform blends the ease of Arduino™'s hardware manipulation with the power of a fully operational Linux operating system. Most sketches (Arduino™ programs) written for Arduino™ boards can be ported over to the Galileo with little or no modification. All popular Arduino™ libraries such as SD, Ethernet, Wi-Fi, EEPROM, SPI and Wire are available along with access to the Linux side of the board with calls via the 'system()' function.



Intel Galileo Board

The Galileo board has the advantage of being able to run Linux environment along with the Arduino™ code, combining the best of both worlds. The Linux

OS can be loaded to an SD card, and Galileo can be set to boot from the Linux image. Using the Linux image, the following features become available:

- ◆ **Wi-Fi drivers:** The Galileo supports Intel Wi-Fi cards via drivers included in the Linux image. There are many mini-PCIe Wi-Fi cards.
- ◆ **Python:** There are Python scripts readily available that can check for unread email or perform tasks. Custom Python scripts can also be easily created to enhance functionality and truly make your device an Internet of Things device.
- ◆ **openCV:** OpenCV or Open Source Computer Vision is a library. Using a webcam connected to the USB port of the Galileo board, a live feed from the camera can be captured to perform some tasks such as object detection or recognition.
- ◆ **SSH:** Secure Shell (SSH) is a command line tool/protocol to securely access a remote computer. This can enable applications such as remote controlling a Galileo that's monitoring a home or simply communicating with the Galileo board without the serial communication port.
- ◆ **Node.js:** Node.js is a library dedicated to building server-side applications in Javascript. Node.js is meant to run on an HTTP server and its applications are event driven. Most suited for web projects.
- ◆ **ALSA:** Advanced Linux Sound Architecture (ALSA) enables the Galileo board to play sound.
- ◆ **V4L2:** Video4Linux2 is a video record and play Linux utility.